

Flashcards — CampusX 100 Days of ML | Day 13–20

23 cards | Front: Question | Back: Answer | Use for Sunday revision

Day 13

Q1

What is the universal 5-step sklearn workflow?

ANSWER

1. Load data. 2. Split X and y. 3. train_test_split. 4. Instantiate model → .fit(X_train, y_train) → .predict(X_test). 5. Evaluate with accuracy_score / metric.

Day 13

Q2

What do test_size and random_state do in train_test_split?

ANSWER

test_size: fraction held out for testing (e.g., 0.2 = 20%).
random_state: seeds the randomizer for reproducible splits — same split every run.

Day 13

Q3

What 3 sklearn methods does every estimator share?

ANSWER

.fit(X_train, y_train) — trains the model. .predict(X_test) — generates predictions. .score(X_test, y_test) — evaluates performance. These work the same for ALL algorithms.

Day 14

Q4

What are the 4 types of data in ML?

ANSWER

Numerical Continuous (height, salary), Numerical Discrete (count, clicks), Categorical Ordinal (rating 1-5, size S/M/L), Categorical Nominal (gender, city). Also: Unstructured (text, images, audio).

Day 14

Q5

What is the difference between structured and unstructured data?

ANSWER

Structured: tabular (rows + columns), can go in SQL. 20% of real-world data. Traditional ML works directly. Unstructured: no fixed format (text, images, audio). 80% of real-world data. Needs DL or feature extraction first.

Day 14

Q6

What is the difference between a Business Metric and an ML Metric?

ANSWER

ML Metric: what the model optimizes — accuracy, F1, RMSE.
Business Metric: what the company cares about — revenue, retention, cost savings. They are NOT the same. Always define both. ML metric is the proxy.

Day 15

Q7

Name 5 important pd.read_csv() parameters and what they do.

ANSWER

sep: column separator. index_col: column to use as index.
usecols: load only specific columns. nrows: read first N rows.
na_values: strings to treat as NaN. encoding: for non-UTF-8 files.

Day 15

Q8

How do you load a large CSV that doesn't fit in RAM?

ANSWER

Use chunksize parameter: for chunk in pd.read_csv('big.csv', chunksize=10000): process(chunk). Each chunk is a DataFrame. Process and discard before loading the next.

Day 16

Q9

How do you read JSON into a pandas DataFrame?

ANSWER

pd.read_json('file.json') for simple flat JSON. For nested JSON: import json; data = json.load(f); then pd.json_normalize(data, record_path=['key']) to flatten nested structure.

Day 16

Q10

What is the SQLite + pandas workflow for reading data?

ANSWER

import sqlite3; conn = sqlite3.connect('db.db'); df = pd.read_sql_query("SELECT * FROM table", conn); conn.close().
To write: df.to_sql('table', conn, if_exists='replace', index=False).

Day 17

Q11

What does an API GET request look like in Python?**ANSWER**

```
import requests; response = requests.get(URL, params={'key': 'value'}); data = response.json(). Check response.status_code == 200 before using. API key usually goes in params or headers.
```

Day 17

Q12

What HTTP status codes should you know?**ANSWER**

200 = OK (success). 400 = Bad Request (wrong params). 401 = Unauthorized (wrong/missing API key). 404 = Not Found (wrong URL). 500 = Server Error. Always check status_code before parsing response.

Day 18

Q13

What is the requests + BeautifulSoup workflow?**ANSWER**

1. requests.get(url) → response.text (HTML string). 2. soup = BeautifulSoup(html, 'html.parser'). 3. soup.find(tag, class_='x') for first match. 4. soup.find_all(tag) for all matches. 5. .text to get content. 6. Build list → pd.DataFrame.

Day 18

Q14

When would you use pd.read_html() instead of BeautifulSoup?**ANSWER**

When the website has proper HTML tags. pd.read_html(url) returns a list of DataFrames for all tables on the page. Much faster than manual scraping. Works well on Wikipedia, financial sites.

Day 19

Q15

What is the difference between mean, median, and mode? When to use each?**ANSWER**

Mean: sum/count. Use for symmetric data with no outliers. Median: middle value. Use for skewed data or data with outliers (salary, house price). Mode: most frequent. Use for categorical data.

Day 19

Q16

What does skewness indicate?**ANSWER**

Asymmetry of distribution. Skew ≈ 0 = symmetric. Positive skew (>0) = right tail, Mean $>$ Median (income, house price). Negative skew (<0) = left tail, Mean $<$ Median. df['col'].skew().

Day 19

Q17

What is IQR and why is it better than range for measuring spread?**ANSWER**

IQR = $Q3 - Q1$ (middle 50% of data). Range = Max - Min. Range is heavily affected by a single outlier. IQR ignores the extremes, giving a robust measure of the typical spread of data.

Day 20

Q18

What does a box plot show?**ANSWER**

$Q1$ (25th percentile) = bottom of box. $Q2$ / Median = line in box. $Q3$ (75th percentile) = top of box. IQR = box width. Whiskers = $Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$. Points outside whiskers = outliers.

Day 20

Q19

What plots do you use for numerical vs categorical univariate analysis?**ANSWER**

Numerical: histogram (distribution shape), boxplot (outliers + spread), KDE plot (smooth distribution). Categorical: countplot / bar chart (frequency per category), value_counts(normalize=True) for proportions.

Day 20

Q20

What 6 things should you look for during univariate EDA?**ANSWER**

1. Distribution shape (normal, skewed, bimodal). 2. Outliers. 3. Missing values pattern. 4. Impossible values (negative age). 5. Cardinality of categoricals. 6. Class imbalance in target variable.

Extra

Q21

What is data leakage and how do you prevent it?**ANSWER**

Fitting a preprocessing transformer (like StandardScaler) on the full dataset before splitting — test set information leaks into training. Prevention: always fit transformers ONLY on X_train, then transform both X_train and X_test.

Extra

Q22

Which algorithms REQUIRE feature scaling and which DON'T?**ANSWER**

Require scaling: KNN, SVM, Neural Networks, Logistic Regression, PCA (distance/gradient based). Don't need scaling: Decision Tree, Random Forest, XGBoost, Naive Bayes (tree-based or probabilistic — not distance based).

Extra

Q23

What is stratified split and when do you use it?**ANSWER**

train_test_split with stratify=y ensures the class proportions in train/test match the original dataset. Use whenever target classes are imbalanced (e.g., 95% negative, 5% positive in fraud detection).

Rule: Cover the answer, read the question, say the answer out loud, then check. Only mark as 'known' if you got it right without peeking.